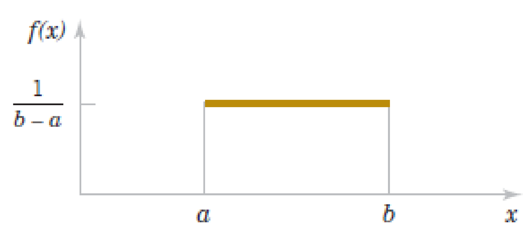
Continuous Random Variables

Families of random variables

▶ **continuous uniform random variables**: if X has the probability density function

$$f(x) = \begin{cases} \frac{1}{b-a}, & \text{for } x \in [a,b] \\ 0, & \text{otherwise} \end{cases}$$

we call X *uniformly distributed*, $X \sim U(a,b)$



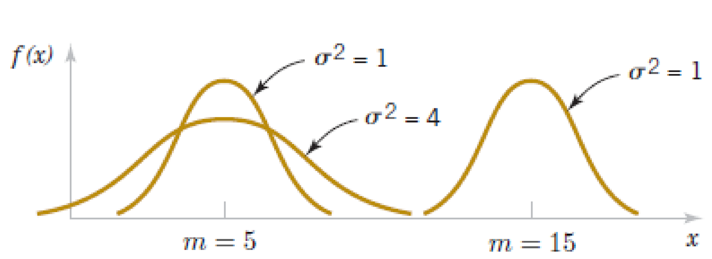
• the mean and the variance are

$$E(X) = \frac{a+b}{2} \qquad var(X) = \frac{(b-a)^2}{12}$$

▶ **normal random variables**: if X has the probability density function

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-m)^2}{2\sigma^2}}$$

we call X *normally distributed*, $X \sim N(m, \sigma^2)$



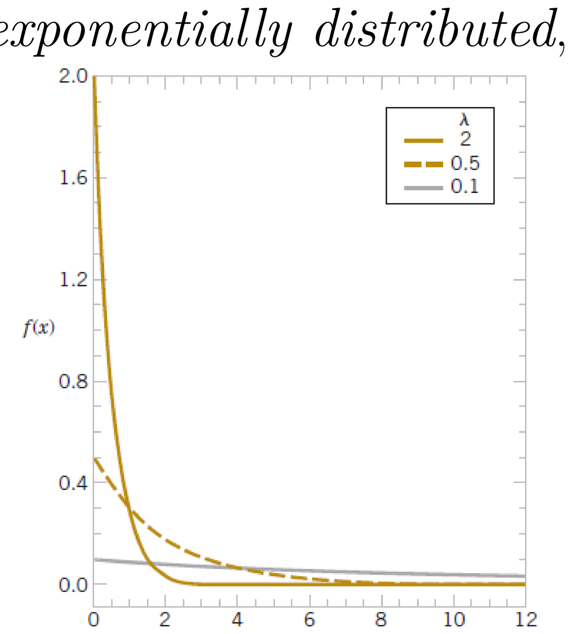
• the mean and the variance are

$$E(X) = m \qquad var(X) = \sigma^2$$

▶ **exponential random variables**: if X has the probability density function

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & , \text{ for } x \geq 0 \\ 0 & , \text{ otherwise} \end{cases}$$

we call X *exponentially distributed*, $X \sim Exp(\lambda)$



• the mean and the variance are

$$E(X) = \frac{1}{\lambda} \qquad var(X) = \frac{1}{\lambda^2}$$

▶ **standard normal random variables**: a variable with a standard normal distribution $Z \sim N(0,1)$ is a normally distributed variable that corresponds to $m = 0$ and $\sigma = 1$

• its cummulative distribution function (CDF) is denoted by

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{t^2}{2}} dt$$

and it has the important values in a z-table like this one

• when working with normally distributed random variables $X \sim N(m, \sigma^2)$ one uses the following **standardization argument**:

$$P(x_1 \leq X \leq x_2) = P\left(\frac{x_1 - m}{\sigma} \leq Z \leq \frac{x_2 - m}{\sigma}\right) = \Phi\left(\frac{x_2 - m}{\sigma}\right) - \Phi\left(\frac{x_1 - m}{\sigma}\right)$$

where $Z = \dfrac{X - m}{\sigma}$ is a standard normally distributed random variable and the corresponding values $\Phi\left(\dfrac{x_2 - m}{\sigma}\right)$, $\Phi\left(\dfrac{x_1 - m}{\sigma}\right)$ can be found in the z-table.

▶ **normal approximations of some discrete random variables**

• when n is large enough, a **binomial random variable** $X \sim Bin(n,p)$ can be approximated by a normally distributed random variable $Y \sim N(np, np(1-p))$

• when X is a **Poisson random variable** with parameter λ and λ is large enough, then X can be approximated by a normally distributed random variable $Y \sim N(\lambda, \lambda)$

• the following *continuity corrections* are recommended

$$P(X = k) \cong P\left(k - \frac{1}{2} < Y < k + \frac{1}{2}\right) \quad \text{and} \quad P(k_1 \leq X \leq k_2) \cong P\left(k_1 - \frac{1}{2} < Y < k_2 + \frac{1}{2}\right)$$

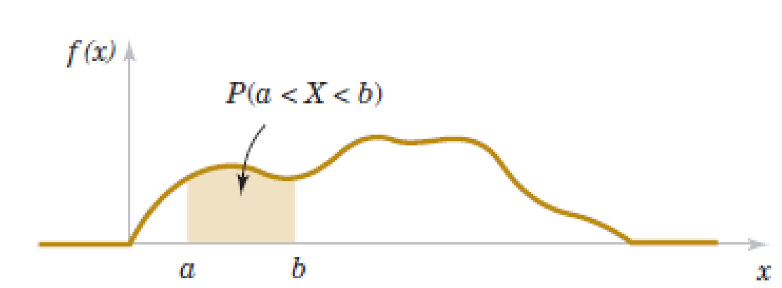
$$P(X \leq k) = P(X < k+1) \cong P\left(Y < k + \frac{1}{2}\right) \quad \text{and} \quad P(X \geq k) = P(X > k-1) \cong P\left(Y > k - \frac{1}{2}\right)$$

Continuous Random Variable

A continuous random variable is a random variable that takes values in an uncountable set, usually an interval. For a discrete random variable the verb was *to count*, now it becomes *to measure*. A continuous random variable measures something.

• X is a continuous random variable when there exists a function $f(x)$, called **probability density function**, such that for all $-\infty \leq a \leq b \leq \infty$

$$P(a < X < b) = \int_a^b f(x)\, dx$$



• a probability density function (PDF) has the defining properties

$$\int_{-\infty}^{\infty} f(x)\, dx = 1 \quad \text{and} \quad f(x) \geq 0$$

• the **cumulative distribution function (CDF)** of a continuous random variable X is

$$F_X(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)\, dt$$

• the **mean and the variance** of a random variable X

$$E(X) = \int_{-\infty}^{\infty} x \cdot f(x)\, dx \qquad var(X) = \int_{-\infty}^{\infty} (x - E(X))^2 \cdot f(x)\, dx$$

Functions of random variables

▶ the CDF of $Y = g(X)$ will be

$$F_Y(y) = P(g(X) \leq y) = P\left(X \leq g^{-1}(y)\right)$$

and the expected value is

$$E[Y] = \int_{-\infty}^{\infty} g(x) \cdot f_X(x)\, dx$$

where f_X is the PDF of X

Probability models of derived r.v.

▶ assume $a > 0$, then for $Y = aX$

$$F_Y(y) = F_X\left(\frac{y}{a}\right) \quad \text{and} \quad f_Y(y) = \frac{1}{a} \cdot f_X\left(\frac{y}{a}\right)$$

$$X \sim U(b,c) \quad \text{then} \quad Y \sim U(ab, ac)$$

$$X \sim Exp(\lambda) \quad \text{then} \quad Y \sim Exp\left(\frac{\lambda}{a}\right)$$

$$X \sim N(m, \sigma^2) \quad \text{then} \quad Y \sim N(am, a^2\sigma^2)$$

▶ if $Y = X + b$, then

$$F_Y(y) = F_X(y - b) \quad \text{and} \quad f_Y(y) = f_X(y - b)$$

Conditional probability models

▶ given an event B, with $P(B) > 0$ the *conditional probability density function* of X is

$$f_{X|B}(x) = \begin{cases} \frac{f_X(x)}{P(B)} & , \text{ if } x \in B \\ 0 & , \text{ otherwise} \end{cases}$$

• the conditional expected value

$$E[X|B] = \int_{-\infty}^{\infty} x \cdot f_{X|B}(x)\, dx$$

• the conditional variance is

$$var(X|B) = E[X^2|B] - E^2[X|B]$$

• a random variable X resulting from an experiment with event space $B_1, B_2, \ldots B_m$ has the PDF

$$f_X(x) = \sum_{i=1}^{m} f_{X|B_i}(x) \cdot P(B_i)$$

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